

Individual Tree Crown (ITC) Delineation, Identification, and Regrouping in High Spatial Resolution (30-100cm/pixel) Aerial and Satellite Multispectral Images

by [François A. Gougeon](#), Ph.D.

Table of contents

- [Research objectives](#)
 - [Individual tree crown delineation](#)
 - [Individual tree crown classification](#)
 - [Regrouping into forest stand or environmental strata](#)
 - [Towards automatic interpretation](#)
 - [References](#)
-

Goal:

The automatic or semi-automatic production of individual tree-based inventories as a goal in itself or as a necessary step in obtaining almost automatically the precise forest inventory parameters needed for sustainable forest development.

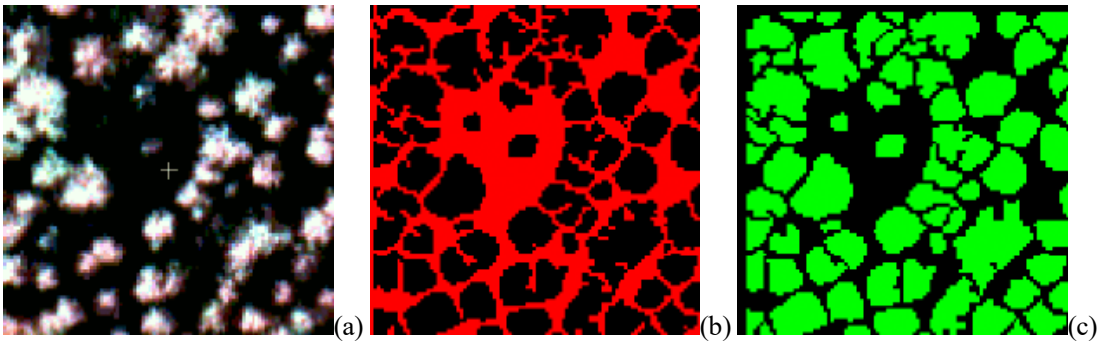
Note: Information on an individual tree basis could be useful to a variety of forestry-related tasks, from management at the landscape level to biodiversity monitoring, as input to computer models (growth, health, ...) and decision support systems, as well as, in the production of criteria and indicators of sustainable forest management success.

Research Objectives:

From high spatial resolution (30-100 cm/pixel) aerial and satellite multispectral images, to develop procedures, methods, and systems to:

- isolate individual tree crowns (ITC) from each other and from the background vegetation;
 - [delineate](#) precisely their crown boundaries;
 - [identify](#) their species; and finally, if needed,
 - [regroup](#) them into forest stands or environmental strata.
-

Individual Tree Crown Delineation: A Valley-Following Approach



A representative section (100 pixels x 100 lines) of:

- a) the original MEIS image (31 cm/pixel)
- b) the initial tree crown isolation by the valley-following program ITCVFOL
- c) the more complete crown delineation by the rule-based program ITCISOL

In one study ([Gougeon, 1995c](#)), 81% of the automatically delineated tree crowns were the same as those obtained by a visual interpretation of the imagery. The automatic tree counts from the image were within 8% of the counts done on the ground.

Individual Tree Crown Multispectral Classification in High Spatial Resolution Images

Various types of multispectral crown-based signatures were analysed for their species recognition capabilities using **manually** delineated tree crowns (i.e., to be independent of the automatic delineation process) ([Gougeon, 1995b](#)). Each classification was repeated ten times to produce a more stable and accurate comparison.

Classification comparison of multispectral signatures
for tree crowns individually delineated on high resolution (36 cm) MEIS images
(with 50 trees/species, 5 species, 10 classification runs/signature type)

Signature types	Mean species accuracies (MSA)					Mean overall accuracy (range)
	BS	JP	RP	WP	WS	
ave_sign	90.6	74.4	54.9	69.3	68.1	71.9 (66.7 - 75.4)
lit_sign	89.8	68.2	66.9	83.7	62.6	74.3 (70.7 - 78.0)
tt_sign	89.5	74.5	59.1	61.7	55.1	67.7 (56.9 - 74.8)
si_sign	85.8	71.3	65.7	80.0	63.4	73.3 (68.3 - 79.1)
mpcl_s	93.8	71.5	57.9	83.5	72.0	76.4 (72.1 - 80.5)
mpclelev_s	95.7	70.0	64.3	80.0	70.3	76.2 (71.2 - 79.2)
mcov_s	89.4	74.1	45.9	71.5	70.3	70.2 (65.1 - 78.1)
Av. MSA	90.7	72.0	59.2	75.7	66.0	72.8
BS = Black Spruce WS = White Spruce						
JP = Jack Pine RP = Red Pine WP = White Pine						

See ([Gougeon, 1995b](#)) for a description of the signature types.

Supervised classification of automatically delineated tree crowns

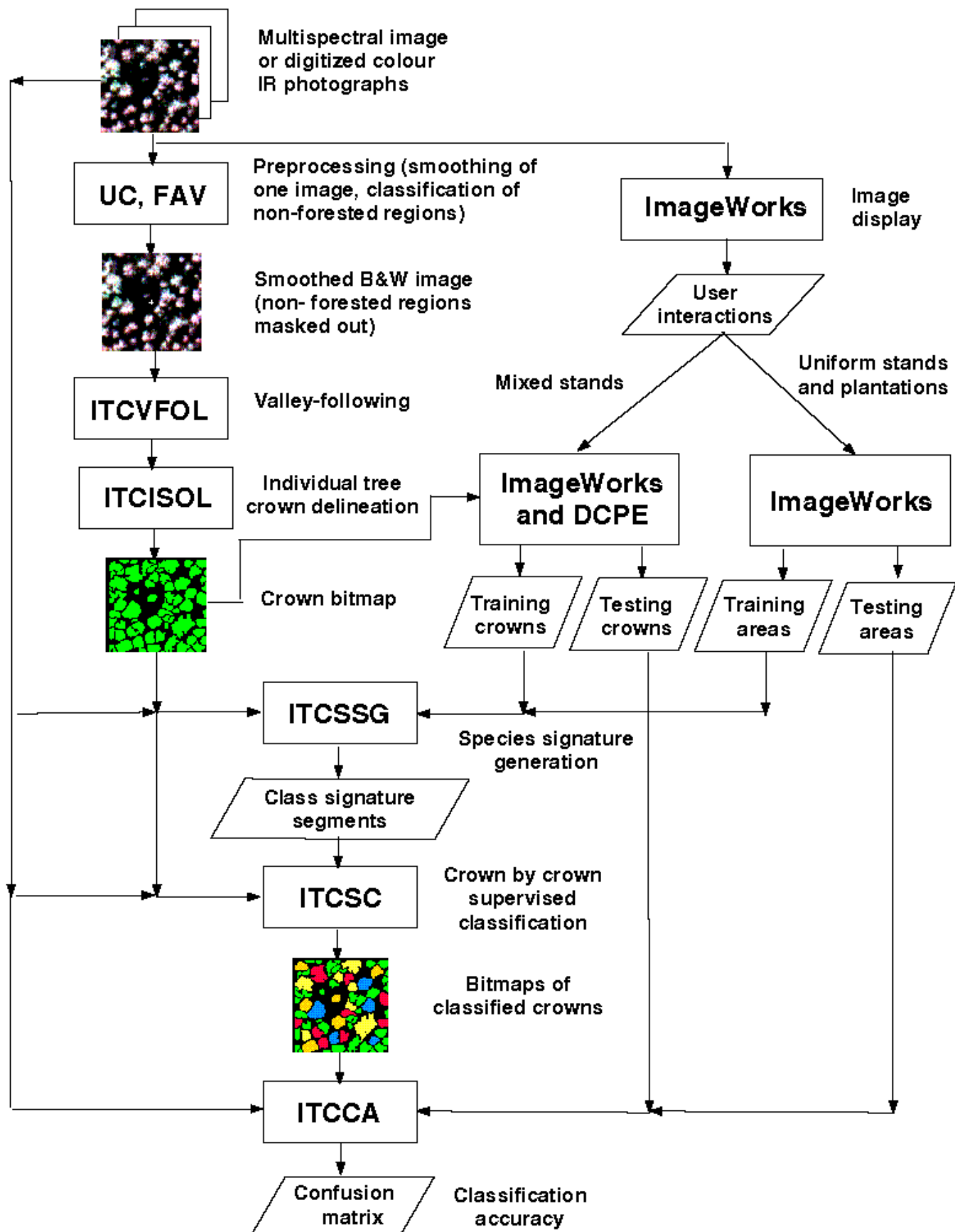
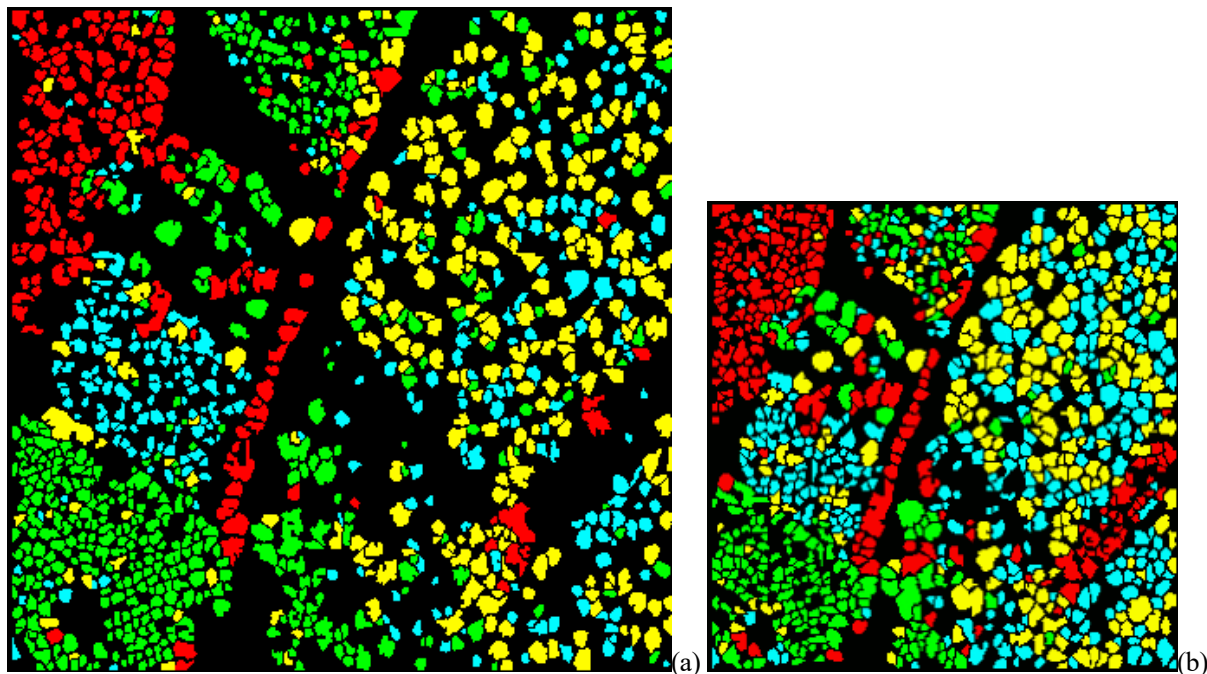


Fig. 1 - Methodology for individual tree crown (ITC) supervised classification

Supervised classification of automatically delineated tree crowns at different spatial resolutions



Multispectral supervised classification into four coniferous species of tree crowns automatically delineated on:

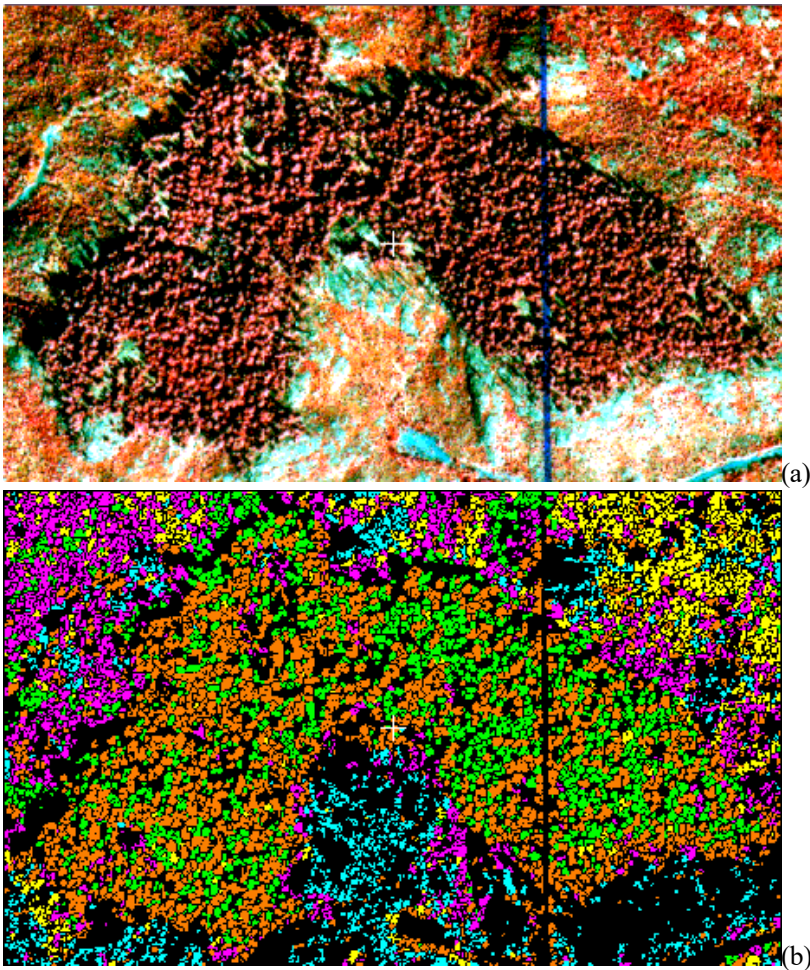
- a 70 cm/pixel multispectral image (a), and
- a 100 cm/pixel multispectral image (b).

Green = Red Pine Blue = Red Spruce
Yellow = White Pine Red = White Spruce

The image used in (b) was generated from the first image (a), simulating "very" approximately an image from the next generation of earth observation satellites (OrbView, Space Imaging, EarthWatch).

Average classification accuracies are typically around 74-81% for four coniferous species, depending on the spatial resolution (36 to 100 cm/pixel) ([Gougeon, 1995a](#)).

Work with digitized colour infrared aerial photography



(a) Colour infrared aerial photograph (1:19 000) of the China Creek region on Vancouver Island, British Columbia, Canada, digitized to a spatial resolution of ~60 cm/pixel.

(b) Isolation, delineation, and classification of the tree crowns detected in image (a) ([Gougeon, 1996](#)).

Note: In collaboration with MacMillan Bloedel Lmtd., Nanaimo, British Columbia, Canada.

Regrouping Tree Crowns into Forest Stands (or Environmental Strata) and Obtaining Detailed Information about their Content

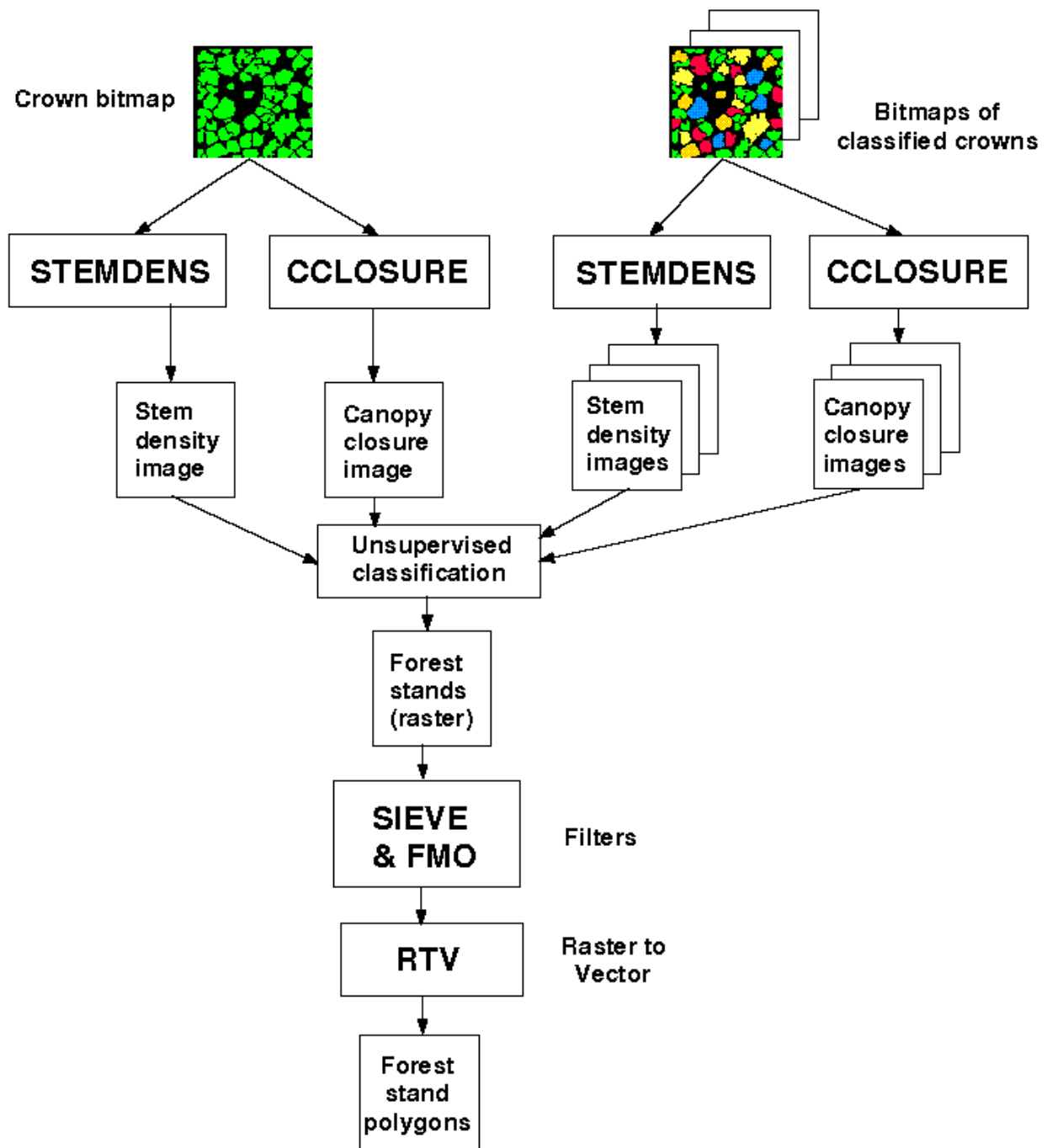
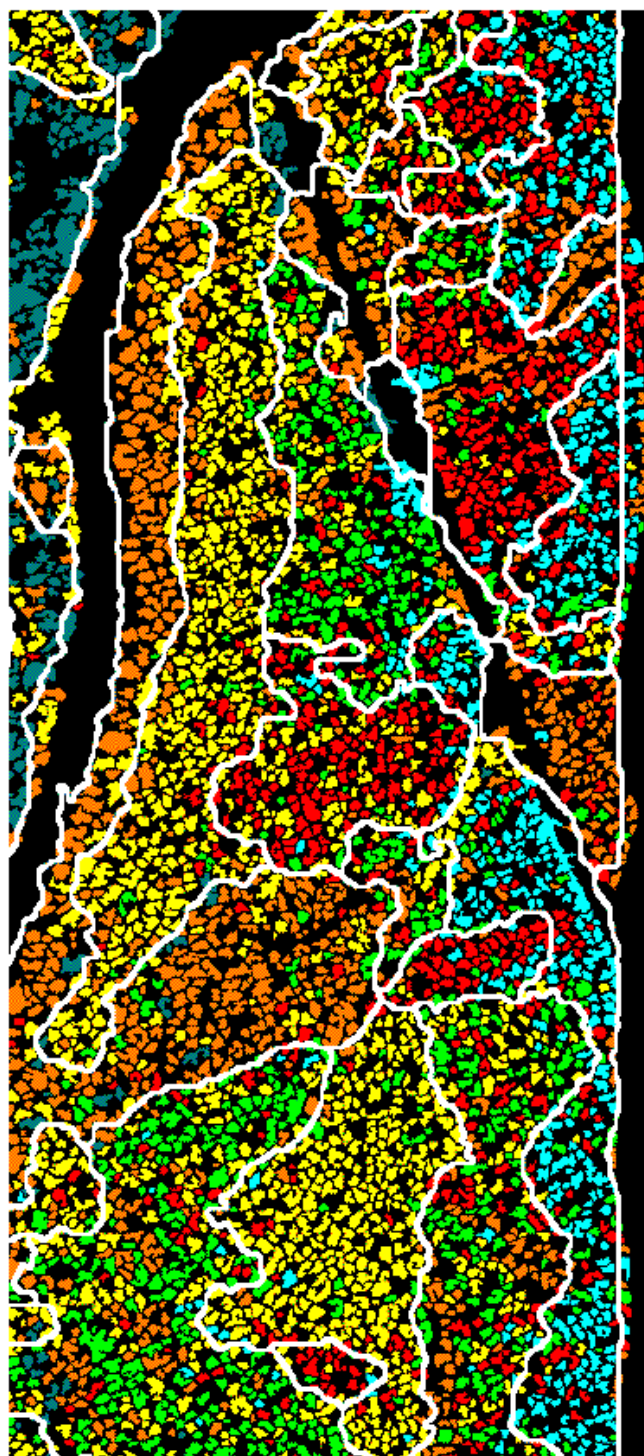


Fig. 2 - Methodology for forest stand delineation from individual tree crown information



(3)



(4)

Fig. 3 - Pseudo-colour infrared rendition of a sub-area of a Casi image (60 cm/pixel) of the Nahmint Lake species trial area in the central forest of Vancouver Island, British Columbia, Canada. The delineated areas (in white) correspond to stands judged relatively uniform by a forester. The training and testing areas used for the classification were delineated where species are known to be homogeneous. Their respective species and colours are: Douglas-fir (*Pseudotsuga menziesii*) in red, grand fir (*Abies grandis*) in green, amabilis fir (*Abies amabilis*) in blue, western redcedar (*Thuja plicata*) in orange, and western hemlock (*Tsuga heterophylla*) in yellow.

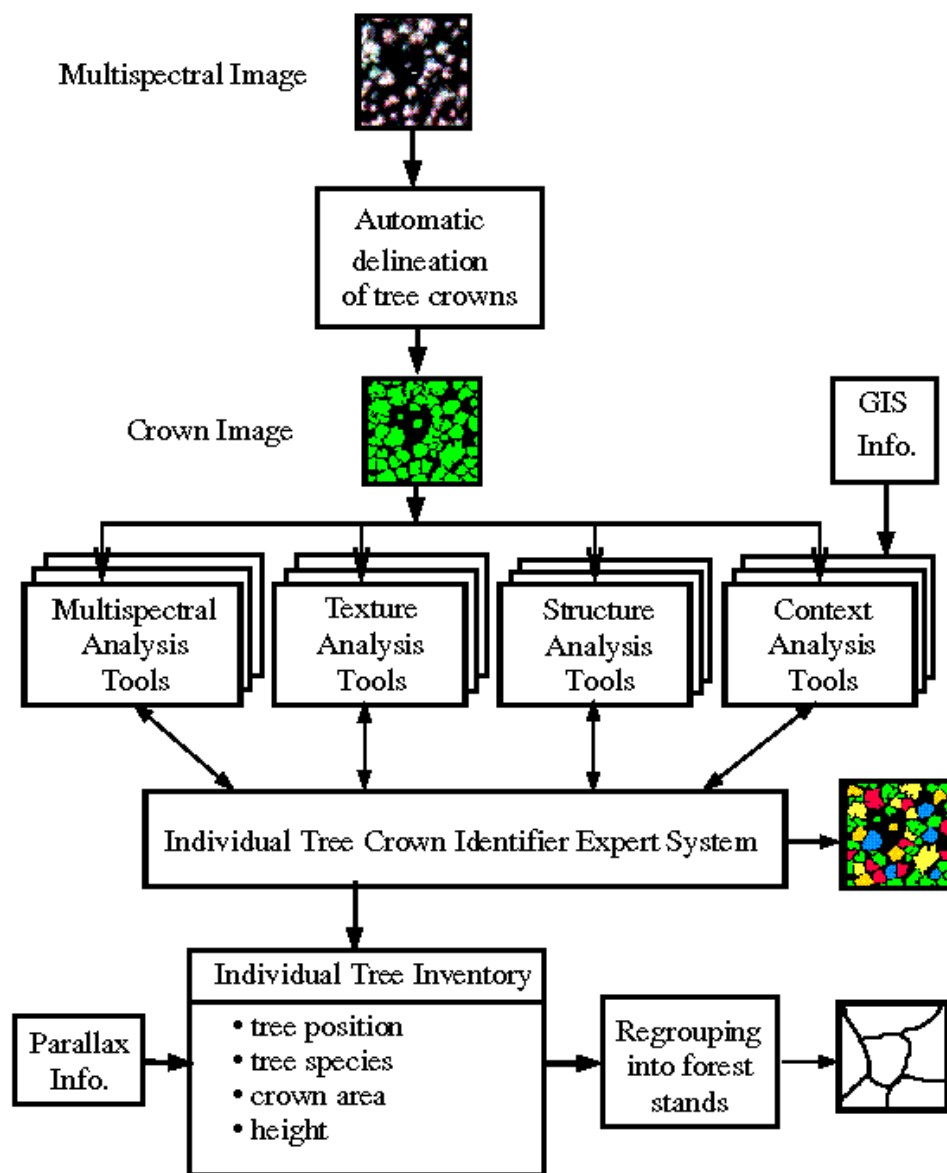
Fig. 4 - The individual tree crown (ITC) classification of Figure 3 and its computer generated forest stand outlines using the same colour scheme as Figure 3.

Transfer of detailed forest information to a Geographic Information System (GIS)



Note: Work was supported in part by funding from Forest Renewal British Columbia (FRBC) awarded to MacMillan Bloedel Ltd., Nanaimo, BC. The Casi data was acquired by Itres Research, Calgary, Alberta, who are collaborating on this FRBC project.

Towards Automatic Interpretation of Aerial Images and High Resolution (one metre/pixel) Satellite Images



**Individual tree based forest inventory from
high spatial resolution images (30 - 100 cm/pixel)**

References:

Please consult the following [bibliography](#). Recent papers are downloadable.

Go to the [table of contents](#) at the top of this file.

Redid page my way (FG) -- Feb. 2026